

Jiwon Kang

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Research Interests

Starting my research career in 3D vision, I expanded my focus during my Master's to Generative AI, practical applications, and Unified Multimodal Models (UMMs). Recently, I am interested in the challenge of world modeling by combining 3D vision capabilities with generative knowledge, which can be extended to real world robotics and immersive gaming environments. Key research areas include:

- Diffusion & Generative AI (Image Generation, Guidance, Sampling)
- Unified Multimodal Models
- 3D Vision (Novel View Synthesis, NeRF, Gaussian Splatting)

Education

M.S. in Artificial Intelligence

Kim Jaechul Graduate School of AI, KAIST

Mar. 2025 – Feb. 2027 (Expected)

Advisor: [Seungryong Kim](#) (CVLAB)

B.S. in Computer Science and Engineering

Korea University

Mar. 2019 – Feb. 2025

- GPA: 4.24 / 4.5
- Double Major in Statistics
- Leave of Absence for R.O.K. Military Service (Dec. 2020 – Jun. 2022)
- Undergraduate Research Intern @ [CVLAB](#) (Sep. 2022 – Feb. 2025), Advisor: Seungryong Kim

Experience

Kakao

Research Intern

Mar. 2026 – May 2026

Seoul, Korea

- Worked on Text-to-Speech research, repurposing Whisper as a continuous tokenizer for autoregressive diffusion-based TTS model.

SAMSUNG

Research Collaboration

Aug. 2026 – Oct. 2026

Seoul, Korea

- Collaborated with Samsung Electronics on background face anonymization via generative face-swapping [\[C3\]](#).

Publications

*: Equal Contribution, †: Corresponding Author

International Conference

[C4] Transferability Between Understanding and Generation in Unified Multimodal Models [Paper](#) | [Project Page](#) | [Code](#)

Jiwon Kang^{*}, Heeji Yoon^{*}, Jaewoo Jung, Jaewon Min, Minkyong Jeon, Biyeon Hwang, Sangwon Jung, Seungryong Kim[†]

- Investigated how understanding and generation capabilities (e.g., counting) transfer to one another within unified multimodal models.
- Showed that transferability depends on the model architecture, and proposed a practical post-training strategy for unified multimodal models that leverages transferability.

European Conference on Computer Vision (ECCV 2026).

[C3] APPLE: Attribute-Preserving Pseudo-Labeling for Diffusion-Based Face Swapping [Paper](#) | [Project Page](#) | [Code](#)

Jiwon Kang^{*}, Yeji Choi^{*}, JoungBin Lee, Wooseok Jang, Jinhyeok Choi, Taekeun Kang, Yongjae Park, Myungin Kim, Seungryong Kim[†]

- Prior diffusion-based face swapping methods focused on changing facial identity, failing to preserve facial attributes.
- Proposed a novel pseudo-labeling framework that preserves facial attributes while swapping identities, achieving Pareto-optimal results.

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026).

[C2] A Noise is Worth Diffusion Guidance [Paper](#) | [Project Page](#) | [Code](#)

Donghoon Ahn^{*}, Jiwon Kang^{*}, Sanghyun Lee, Jaewon Min, Wooseok Jang, Minjae Kim, Hyungwon Cho, Sayak Paul, SeonHwa Kim, Eunju Cha[†], Kyong Hwan Jin[†], Seungryong Kim[†]

- Showed that classifier-free guidance can be replaced by refining the initial noise.
- Enabled high-quality generation without a guidance pass, highlighting the importance of the initial noise in diffusion models.

International Conference on Learning Representations (ICLR 2026).

[C1] Where and How to Perturb: On the Design of Perturbation Guidance in Diffusion and Flow Models [Paper](#) | [Project Page](#) | [Code](#)

Donghoon Ahn^{*}, Jiwon Kang^{*}, Sanghyun Lee, Minjae Kim, Wooseok Jang, Jaewon Min, Sangwu Lee, Sayak Paul, Seungryong Kim[†]

- Systematically studied perturbation guidance in diffusion and flow models.
- Showed that perturbing different layers/heads in a diffusion model exhibits different effects on the generated samples, which can be systematically combined to improve user-specified attributes via the proposed framework.

Conference on Neural Information Processing Systems (NeurIPS 2025).

Workshop

[W1] Self-Evolving Neural Radiance Fields [Paper](#) | [Project Page](#) | [Code](#)

Jaewoo Jung^{*}, Jisang Han^{*}, Jiwon Kang^{*}, Seongchan Kim, Min-Seop Kwak, Seungryong Kim[†]

- Proposed a self-training framework that lets a NeRF improve itself from few input views via pseudo-label distillation.

ICCV 2025 Workshop on Wild3D: 3D Modeling, Reconstruction, and Generation in the Wild.

Preprints & Under Review

[P1] RAIN-GS: Relaxing Accurate Initialization Constraint for 3D Gaussian Splatting [Paper](#) | [Project Page](#) | [Code](#)

Jaewoo Jung^{*}, Jisang Han^{*}, Honggyu An^{*}, Jiwon Kang^{*}, Seonghoon Park^{*}, Seungryong Kim[†]

- Relaxed the accurate point-cloud initialization requirement of 3D Gaussian Splatting, enabling training from sparse or random initialization.

Under Review.

Extracurricular Activities

Military Service (KATUSA, Korean Augmentation to the U.S. Army) Dec. 2020 – Jun. 2022

- MOS : 92Y (Unit Supply Specialist)
- 194th DSSB, 2nd Infantry Division, Eighth U.S. Army, Camp Humphreys

AIKU (Artificial Intelligence Society of Korea University) Sep. 2022 – Aug. 2023

- Co-founder & Vice President of academic research society for undergraduate students